

MOCABBAN : a modeling case base by a bayesian network applied to the diagnosis of hepatic pathologies

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Abstract

Our work presents a case based reasoning system for the diagnosis of hepatic pathologies according to a bayesian model. The main idea consists a modelling the case base by a bayesian network. Bayesian networks are excellent tools for the modelling of the uncertain in terms of their clear graphic representation as well as the conditional probabilities laws defined on this graph. Our network allows a representation of a qualitative and causal knowledge in addition to a quantitative knowledge which expresses the uncertainty. It contains four levels: clinical, biologic, medical imaging and diagnosis-therapy. Each level has a table containing the different conditional probabilities in order to get a good final diagnosis. Each level is composed of a set of attributes; each attribute corresponds to a node of a network. The arcs describe the relations between these attributes as conditional probabilities of attributes in the case. We have used the exact algorithm of inference JLO-Jensen, Lauritzen et Olesen- to calculate the conditional probabilities. When we have a new case, it is inserted in the network and is propagated to the network. When some signs are missed, we have used another algorithm of learning EM- Expectation Maximisation to estimate the conditional probabilities of the missed variables in order to get a good learning. We have chosen this modelling in order to obtain a good final diagnosis.

1. Introduction

In order to improve the performances of the system CBR. The reasoning use cases already stocked in a case base. The later is supposed to be representative of the set of problems susceptible to be put in the system. But more the base develops, more the time of computing will be long. This is why the techniques of organizations of memory and the algorithms of the

search and the appraisalment are particularly important. In [1] we can find several organizations. The domain of our work is medicine where the uncertainly [2] is a permanent companion in medical activity.

The probabilistic models, such as Bayesian networks are largely used in the domains based on decision [3]. These domains represent an uncertain knowledge of the treated domain and they are adaptable according to the variability of observations.

Our approach is based on the modelization of the case base using a Bayesian network. It offers a frame of study with various possibilities. Concerning our problem, knowing the diagnosis of hepatic pathologies, the use of Bayesian networks is a good help.

This modelization is based on the description, using graphs, of the relations of causality existing between the variables the define the domain of illnesses of the liver. We associate to each variable a distribution of probability. To calculate the conditional probabilities, we have used the two algorithms of learning JLO and EM to have a good learning. These probabilities are used as a measure of similarity, choosing the case, selecting the case the most probable. In addition, this formalization of the memory of case offers a lot of possibilities in order to obtain a CBR system with a reduced computing time and that produces a good diagnosis.

Under this article, we briefly introduce the case based reasoning in section 2, then the inference in the bayesian network in section 3. Section 4 is devoted to the general description of our medical problem. Section 5 presents the implementation of the bayesian network in CBR system that helps in the diagnosis of hepatic pathologies. The description of a case from the system is presented in section 6. Section 7 shows the detailed architecture of MOCABBAN (**M**odeling **C**ase **B**ase by **B**ayesian **N**etwork) system. At the end there is a conclusion and some perspectives.

2. Case Based Reasoning

2.1. General principle of CBR

In general, the case base reasoning (CBR) is an approach of problem solving which is based on the reuse, by analogy, of past experiences called *case* [7]. a case especially means a problem and the solution applied to solve it (or a method which permits to generate this solution). The reasoning usually consists of four main phases [2] :

- *Retrieve* phase in the beginning, this phase search for similar cases to the current problem;
- *Reuse* phase it finds a solution to the current problem which is based on cases identified in the previous phase;
- Phase of *Revising* the solution which leads to its refinement through evaluation.
- *Learning* phase it permits to update the elements of reasoning taking into account the experience just happened, which can be used in the future[3].

2.2. Base of case

The base of case is a set of solutions of the same problem. It is very important to settle a good CBR. The problem of representing cases always with the choice of the information we want to save, finding the adequate structure, organization of cases and indexing [3] [11]. In our case, the problem of diagnosis, the observations on which the medical decision is based are imperfect and of great quality, uncertain. Thus, The diagnostic therapeutic or prognostic decisional process is under uncertainty. For this reason, we should model knowledge by a bayesian network, in the diagnostic of hepatic pathologies.

3. Inference in the bayesian network

3.1. Bayesian network

A bayesian network (probabilistic or bayesian networks BNs) is a model which shows uncertain knowledge in a complicated phenomenon and permits a real reasoning according to data [9]. It enables a representation of a set of random access variables whose we know a number of dependence relations. The set of variables is called U and the distribution of probabilities in this set is $P(U)$. If we get a new information about one or many variables we will wish to reupdate the knowledge which represents the bayesian network through $P(U)$ according to this new

information. This updating process got using Bayes rule is called inference. Mathematically speaking, inference in a bayesian network is the calculation of $P(U|x)$, i.e, the calculation of the posteriori probability *in* of the network knowing x [8] [9].

In the case of our problem, the observations on which the hepatic diagnosis of pathologies is based are imperfect. They may be uncertain because of that the clinical knowledge is the expression of statistic observations on sample patients suffering from frequent illnesses which have different clinical forms and sometimes do not show the same symptomatology, similar in certain signs with other illnesses or giving an inflected answers to a given treatment. Thus, the decisional process of diagnosis is under uncertainty.

3.2 Inference

3.2.1. JLO algorithm : The Junction Tree

The main use of bayesian network is in calculating the events conditional probabilities related to eachother by causal relationships to that effect which we call *inference*.

The graph consists of n noted nodes $X=X_1, X_2, \dots, X_n$. The inference consists on calculating:

$$P(X_1 | Y) \text{ when } Y \subset X, X \neq Y$$

Y is a set of variables or observation vectors noted O .

We may use exact or approximate algorithms to calculate the conditional probabilities. The most used one is the joining tree algorithm JLO. It consists of transforming the initial network to a new structure called joining tree. We must follow these steps to get this tree[9][12]:

moralization: by joining the parents to eachother and canceling the directions.

Triangularization: by adding arcs selectively to the moral graph (so that we avoid getting cycles ordered in four or more).

Joining tree: is got from the triangularized graph by connecting the cliques in such a way that all the cliques between X and Y contain $X \cap Y$.

We remind a clique is a complete maximal under graph :

- Complete: each node pair (variables) is distinct and joined by an arc.
- Maximal: the clique is not strictly contained in a complete under graph.

The figure 1 represents an example of the Bayesian network that modelize in a reasonable manner the process of the diagnosis of two illnesses of the liver. The mode of reasoning in here presented with a

diagram of causality. The probabilities are determined by the expert and as tables of probabilities necessities a lot of computing, we have used an exact algorithm of inference this transforms our initial graph to a tree of junction. The probabilities are calculated following this tree.

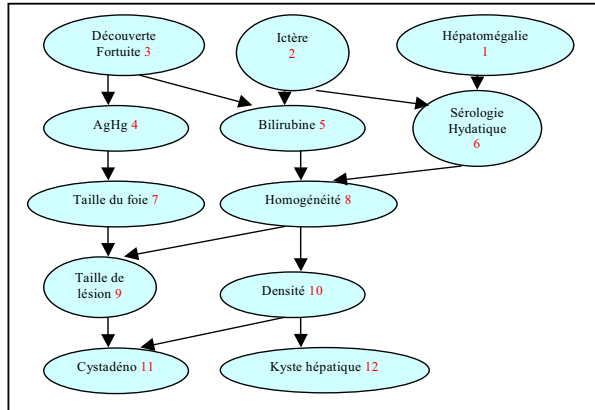


Figure 1. Example of bayesian network with two pathologies of the liver

3.2.2. The EM algorithm

The principal of the algorithm EM is to treat the bases of incomplete examples without adding a new modality (variable non measured) in each node. This iterative algorithm part of an initial structure to estimate the distribution of probability of the missed variables thanks to the classic algorithm EM. The hope of the rate with regard to this variables is then calculated for all the Bayesian networks of the neighborhood to chooses the following structure[3][11].

4. General description of medical problem

The medical problem considered in this article is detecting hepatic pathologies. The medical reasoning starts from a clinical examination and a set of questioning data. In fact, the doctor will identify the predisposed factors basing on the observed symptoms, for instance pains, hypertrophia, icterus ... to establish a number of hypotheses. Knowing that no clinical

examination is specific for 100 %, the uncertainty remains anyway. So the doctor makes its final diagnosis only through the radiological examination. Figure (1) schematizes the different steps of a diagnosis.

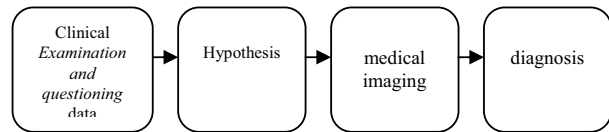


Figure 2. Description of the diagnosis problem

5. Implementation of bayesian network in CBR help system of hepatic diagnosis

The construction of our base of case consists of three distinct steps:

- Qualitative step: only the influence relationships which exist between the variables are taken into account.
- Probabilistic step: in this stage, we should introduce the idea of a clause distribution of the variables and we adjust it according to the created graph.
- Quantitative step: it simply consists of specifying numerically the distributions of conditional probabilities.

5.1. Network variables definitions

In order to use the formalism of bayesian networks, we have deliberately shown the useful information for our problem representation. After discussions with experts in the field, we have deduced that the most relevant information to be treated are divided into four levels:

- Clinical level: It consists of a set of possible factors for detecting the different hepatic illnesses. For example; alcohol, tobacco, dietetics, arterial high blood pressure (HTA), vascular abnormality, cancerological antecedent ...
- Hypothesis level: defines the pathology phase. It shows all the possible pathology cases which may be detected from the previous level. For instance;

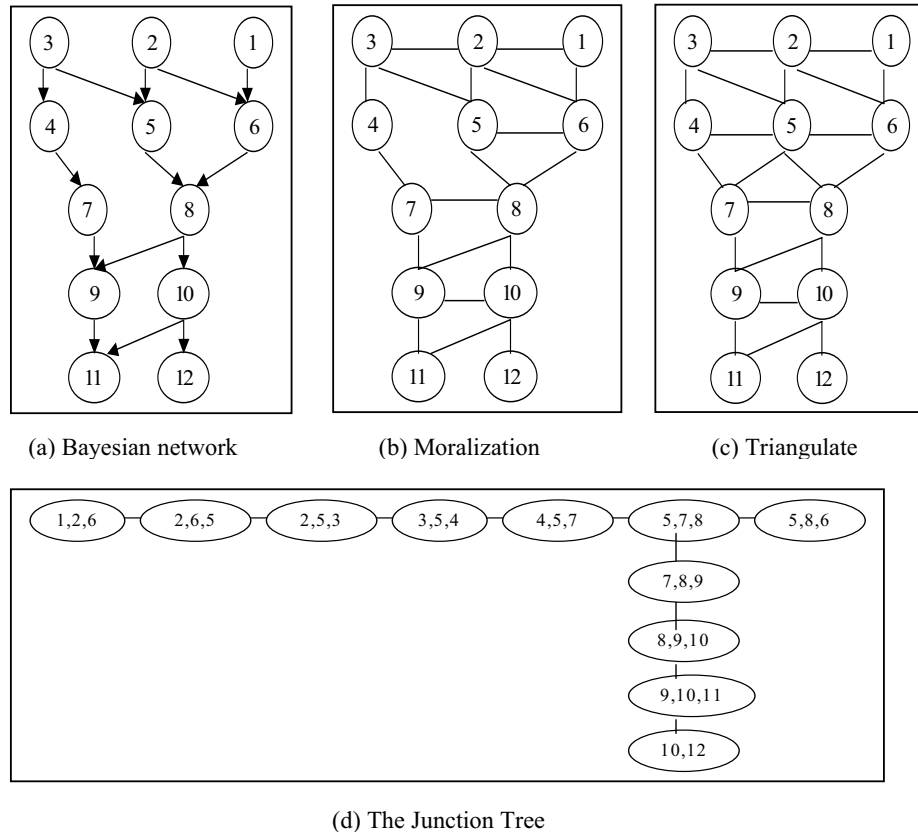


Figure 2. An example of construction of the Junction tree from a bayesian network with two pathologies of the liver

- cirrhosis, hepatic cell carcinoma, hepatic metastasis, lymphome (II) of liver, ...
- medical imaging level: contains the ultrasounded scan and scanning images of liver.
- Diagnosis and therapy level: it shows the final diagnosis; hepatic cyst, hepatic metastasis abscess ...

These information are presented as discrete values variables, each variable is associated with a specific probability and a matrix which shows the assessment of the conditional probabilities.

5.2. Network structure modeling : the base of case

The work described in this article aims at constructing a bayesian network for modeling the base of case which consists of a set of information for the diagnosis. The network nodes represented as the variables of the four levels describe the dependence between the nodes, shown in the previous paragraph. Figure 2 shows the first step of modeling qualitatively the problem and determines the influence between the variables.

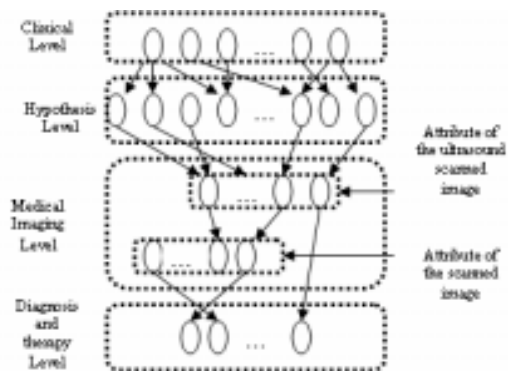


Figure 3. modeling of the base of case by a bayesian network

6. Description of a case in the system

The case is described in terms of the attribute dimension. These attributes are clinical data, hypotheses, characteristics extracted from the image of

liver and the concluded diagnosis. Figure (3) shows a sample of a case from our system.

Description of a case	
Clinical part:	
Pain = 1	
Icterus = 0	
Tobacco = 1	
Medical imaging	
Attribute of the ultrasound scanned	
Size = ...	
Homogeneity = ...	
Attribute of the scanned image	
Size = 620	
Density = 125	
Acute angle = ...	
Homogeneity = ...	
Solution (diagnosis)	
cirrhosis	

Figure 4. case description of MOCABBAN system

7. Detailed architecture of mocabban system

7.1. Case base

The base of case of this system (figure 4) is a set of case vectors, noted as $C = \{ (c_1, p_1), (c_2, p_2), \dots, (c_n, p_n) \}$, each case is presented in two vectors C_k and P_k , as in $C_k = (C_{k1}, C_{k2}, \dots, C_{kn})$, $C_{ki} \in \mathbb{R}$ is a combination of numerical values associated with each case attribute K and $P_k = (P_{k1}, P_{k2}, \dots, P_{kn})$, $P_{ki} \in [0.1]$ is a probability vector which contains probabilities approximate to the case attribute.

MOCABBAN system consists of a sub-system of image treatment. This sub-system works through many phases. A phase of pretreatment, acquisition and numerization of liver image. It aims at getting rid of a great deal of noise. A segmentation phase which aims at separating the elements, existing in the image, from each other. The analysis phase aims at extracting the index (size, perimeter, density, ...) representing the liver image.

The system needs in input the 18 clinic lines of new case. After entry and validation of this information, the information are propagated in the network to obtain the posteriori probabilities of the next level using the algorithm JLO. If the system detects a probability that is superior to 1.0, it predict the user by posting the message "impossible to go on". This network facilitates the search for the cases similar to the case of entry through the obtained conditional probabilities using the node of the diagnostic level the most probable. The architecture of the MOCABBAN is composed of 4 functions: the Bayesian inference

(learning), the research, the adaptation and the enrichment.

7.2. Bayesian inference

7.2.1. The junction tree

the junction tree consists of reducing the network by fusion certain nodes of the graph. For this purpose are must go through steps. The first step, the moralization, the triangulate, add a rope to cycle to obtain of tree element or more. At the end of these two steps all the maximum cliques are fusion to constitute a tree.

7.2.2. Algorithm EM

if some signs are missed, our system uses algorithm of learning EM to estimate the conditional probabilities. This algorithm goes through two steps:

- estimation of the missed variables with actual parameters (calculation of $P(H|V)$ such as V known variables, H unknown variables).
- The use of known variables to estimate new parameters for the Bayesian network.

7.3. Research

The research is done according to the likeness between the new case and the known cases in the base. Let C the new case described by a set of observations $X = \{X_1, X_2, \dots, X_n\}$, that are clinical, biological signs and the attributes of the medical imagery of the liver. This phase consists of determining the diagnosis that is associated to this new case basing on the probabilities (obtained by the Junction tree) as measures of similarity.

From these observations $X_i = c_i$, we look for maximizing :

$$\arg \text{Max} (P(C=c_i | e))$$

C : is a random variable that may take a lot of values corresponding to the set of cliques c_i . According to the algorithm of JLO, for the Junction tree c_i :

$$P(e) = \frac{\prod_{c \in C} \Phi_c(X_c)}{\prod_{s \in S} \Phi_s(X_s)}$$

where Φ_c (resp Φ_s) is potential function of the clique C (resp separation S).

7.4. Adaptation

The adaptation of our system MOCABBAN seems to be good for the improvement of the system quality.

This adaptation is done with the help of the expert of domain. So, the doctor will be able to give an attitude non quantitative to the diagnosis given to the system. But this quantitative attitude will help in this context to estimate the parameter of the Bayesian network, that is to say, in order to provide a more correct diagnosis, and closer to the way the doctor thinks.

7.5. Enrichment

After the judgment of the expert of the domain has given this attitude, if the case seems to be useful to the system, it must be saved in the base, even if it presents a new pathology that must need a specific treatment. This help to enrich of our system with other new information.

We have used a dynamic Bayesian network each time a new information is obtained, nether let it corresponds a new pathology or signs with regard to this pathology, it will be insert in the network and will be divided to the set of network. Before a phase of propagation, JLO permit inserts the maximum obvious variables.

The enrichment is a process that will be realized as follow:

The observed characteristics from the new case are entred in the Bayesian network (the Junction tree as evidences). The Junction tree is updated that is to say the inference method is applied as follow:

- a) for each clique containing the variable v (the sign present in the new case) receiving the evidence (an evidence is a function $\varepsilon : X \in [0,1]$), the potential function of the clique ϕ_c (that is to say its table of probabilities) is multiplied by the evidence.

After entry of one or more a lot of information in the network, the table of probabilities of the clique of the Junction tree as follow:

$$P(X|\varepsilon) = \frac{\prod_{c \in C} P(X_c|\varepsilon)}{\prod_{s \in S} P(X_s|\varepsilon)}$$

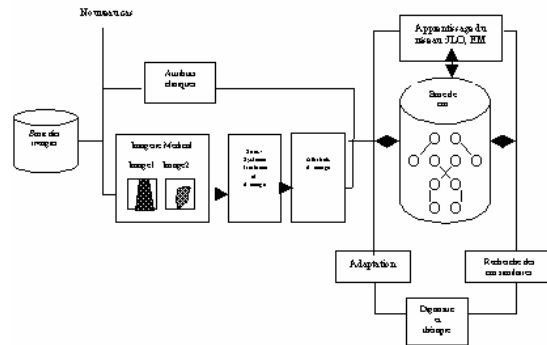


Figure 5. detailed architecture of MOCABBAN System

Figure 6 summarize our system MOCABBAN (Modeling case based of CBR system by Bayesian network) in the form of algorithm.

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BN : Réseau Bayésien
L : Bibliothèque de cas
NC : Nouveau cas
MOCABBAN (BN, L, NC)
DEBUT
  Algorithme JLO (BN, Tree); // construire l'arbre de Jonction
  SI il existe des signes manquants dans le nouveau cas ALORS algorithme
  EM(BN,NC) ; // on applique l'algorithme EM pour calculer les probabilités des
  variables manquantes
  Propagation (NC, Tree) ;
  A ← Recherche (NC, Tree) ;
  C ← max(P(D|NC)) ; // D est le diagnostic associé au nouveau cas
  SI échec alors DEBUT
    Adaptation (NC, Bn, L, Tree) ;
    Enrichir(NC, Tree) ;
  FIN ;
Fin.

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Figure 6. MOCABBAN: Modeling Case Based of CBR system by Bayesian Network

8. Conclusion and perspectives

The work presented in this article is the conception of a system which helps the radiologist doctors. Our problem solving methodology works in CBR paradigm which is based on a bayesian model for modeling the knowledge of the base of the case. The structure of this model is knowledge which is useful for the diagnosis. This knowledge is uncertain. It is divided into four levels. Each level has a definite probability. The aim of our system is to get the best diagnosis.

The Bayesian structure of the case base facilities the research cases that are similar to the case of entry. By choosing the case the most probable. Such a structure

simplifies the hardware of the base in order to insert a new case. The adaptation is done through adaptation of the associated probabilities to the problem in question in entry with the help of the expert in order to solve the new case.

The objective of our future work will be the experimental realization of our Bayesian system CBR, and comparing the results with regard to system DIAPH [13]: a case based reasoning system to help the diagnosis of hepatic pathologies that we have using 55 cases. It is based on asset of clinical symptomatology and of necessary characteristics extracted from the image of the liver the diagnosis of illnesses of the liver. For of the research of similar cases, we have used two measures of similarity Block City and Euclidien that have given to us a rate of similarity between 98% and 99.98%.

Among the perspectives envisageables from our system we have:

- How shall we adapt the network parameters having only a qualitative frame of reference as an advice or a wish expressed by the doctor ?
- How shall we adapt the network structure in an efficient way without recalculating the set of probabilities of data rather than the new data which are available, assuring the validity of our model ?

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